RIGHT EXACT FUNCTORS ON FINITELY PRESENTED MODULES ARE EQUIVALENT TO PAIRS (K, K: A → END_B(K))

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 0GMN: sources/stacks-chow/tags.txt, source lines 20763–20763. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A be a ring. Let $\mathcal{B}$ be an additive category with cokernels. There is an equivalence of categories between*

1. *the category of functors $F : Mod_A^{fp} \to \mathcal{B}$ which are right exact, and*
2. *the category of pairs (K, κ) where $K \in \mathrm{Ob}(\mathcal{B})$ and $\kappa : A \to End_{\mathcal{B}}(K)$ is a ring homomorphism*

given by the rule sending F to $F(A)$ with its natural A-action.

Proof. Let (K, κ) be as in (2). We will construct a functor $F : \mathrm{Mod}_A^{fp} \to \mathcal{B}$ such that $F(A) = K$ endowed with the given A-action κ. Namely, given an integer $n \geq 0$ let us set

$$F(A^{\oplus n}) = K^{\oplus n}$$

Given an A-linear map $\varphi : A^{\oplus m} \to A^{\oplus n}$ with matrix $(a_{ij}) \in \mathrm{Mat}(n \times m, A)$ we define

$$F(\varphi) : F(A^{\oplus m}) = K^{\oplus m} \longrightarrow K^{\oplus n} = F(A^{\oplus n})$$

to be the map with matrix $(\kappa(a_{ij}))$. This defines an additive functor F from the full subcategory of Mod_A^{fp} with objects 0, A, $A^{\oplus 2}$, $\ldots$ to $\mathcal{B}$; we omit the verification.

For each object M of Mod_A^{fp} choose a presentation

$$A^{\oplus m_M} \xrightarrow{\varphi_M} A^{\oplus n_M} \to M \to 0$$

of M as an A-module. Let us use the trivial presentation $0 \to A^{\oplus n} \xrightarrow{1} A^{\oplus n} \to 0$ if $M = A^{\oplus n}$ (this isn't necessary but simplifies the exposition). For each morphism

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$f : M \to N$ of Mod_A^{fp} we can choose a commutative diagram

$$\begin{array}{ccccccc} A^{\oplus m_M} & \xrightarrow{\varphi_M} & A^{\oplus n_M} & \longrightarrow & M & \longrightarrow & 0 \\ \downarrow{\scriptstyle \psi_f} & & \downarrow{\scriptstyle \chi_f} & & \downarrow{\scriptstyle f} & & \\ A^{\oplus m_N} & \xrightarrow{\varphi_N} & A^{\oplus n_N} & \longrightarrow & N & \longrightarrow & 0 \end{array} \tag{0.1.1}$$

Having made these choices we can define: for an object M of Mod_A^{fp} we set

$$F(M) = \text{Coker}(F(\varphi_M) : F(A^{\oplus m_M}) \to F(A^{\oplus n_M}))$$

and for a morphism $f : M \to N$ of Mod_A^{fp} we set

$$F(f) = \text{the map } F(M) \to F(N) \text{ induced by } F(\psi_f) \text{ and } F(\chi_f) \text{ on cokernels}$$

Note that this rule extends the given functor F on the full subcategory consisting of the free modules $A^{\oplus n}$. We still have to show that F is a functor, that F is additive, and that F is right exact.

Let $f : M \to N$ be a morphism Mod_A^{fp}. We claim that the map $F(f)$ defined above is independent of the choices of ψ_f and χ_f in (0.1.1). Namely, say

$$\begin{array}{ccccccc} A^{\oplus m_M} & \xrightarrow{\varphi_M} & A^{\oplus n_M} & \longrightarrow & M & \longrightarrow & 0 \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \chi} & & \downarrow{\scriptstyle f} & & \\ A^{\oplus m_N} & \xrightarrow{\varphi_N} & A^{\oplus n_N} & \longrightarrow & N & \longrightarrow & 0 \end{array}$$

is also commutative. Denote $F(f)' : F(M) \to F(N)$ the map induced by $F(\psi)$ and $F(\chi)$. Looking at the commutative diagrams, by elementary commutative algebra there exists a map $\omega : A^{\oplus n_M} \to A^{\oplus m_N}$ such that $\chi = \chi_f + \varphi_N \circ \omega$. Applying F we find that $F(\chi) = F(\chi_f) + F(\varphi_N) \circ F(\omega)$. As $F(N)$ is the cokernel of $F(\varphi_N)$ we find that the map $F(A^{\oplus n_M}) \to F(M)$ equalizes $F(f)$ and $F(f)'$. Since a cokernel is an epimorphism, we conclude that $F(f) = F(f)'$.

Let us prove F is a functor. First, observe that $F(\text{id}_M) = \text{id}_{F(M)}$ because we may pick the identities for ψ_f and χ_f in the diagram above in case $f = \text{id}_M$. Second, suppose we have $f : M \to N$ and $g : L \to M$. Then we see that $\psi = \psi_f \circ \psi_g$ and $\chi = \chi_f \circ \chi_g$ fit into (0.1.1) for $f \circ g$. Hence these induce the correct map which exactly says that $F(f) \circ F(g) = F(f \circ g)$.

Let us prove that F is additive. Namely, suppose we have $f, g : M \to N$. Then we see that $\psi = \psi_f + \psi_g$ and $\chi = \chi_f + \chi_g$ fit into (0.1.1) for $f + g$. Hence these induce the correct map which exactly says that $F(f) + F(g) = F(f + g)$.

Finally, let us prove that F is right exact. It suffices to show that F commutes with coequalizers, see Categories, Lemma 0GMN. For this, it suffices to prove that F commutes with cokernels. Let $K \to L \to M \to 0$ be an exact sequence of A-modules with K, L, M finitely presented. Since F is an additive functor, this certainly gives a complex

$$F(K) \to F(L) \to F(M) \to 0$$

and we have to show that the second arrow is the cokernel of the first in $\mathcal{B}$. In any case, we obtain a map $\text{Coker}(F(K) \to F(L)) \to F(M)$. By elementary commutative algebra there exists a commutative diagram

$$\begin{array}{ccccccc} A^{\oplus m_M} & \xrightarrow{\varphi_M} & A^{\oplus n_M} & \longrightarrow & M & \longrightarrow & 0 \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \chi} & & \downarrow{\scriptstyle 1} & & \\ K & \longrightarrow & L & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

Applying F to this diagram and using the construction of $F(M)$ as the cokernel of $F(\varphi_M)$ we find there exists a map $F(M) \to \mathrm{Coker}(F(K) \to F(L))$ which is a right inverse to the map $\mathrm{Coker}(F(K) \to F(L)) \to F(M)$. This first implies that $F(L) \to F(M)$ is an epimorphism always. Next, the above shows we have

$$\mathrm{Coker}(F(K) \to F(L)) = F(M) \oplus E$$

where the direct sum decomposition is compatible with both $F(M) \to \mathrm{Coker}(F(K) \to F(L))$ and $\mathrm{Coker}(F(K) \to F(L)) \to F(M)$. However, then the epimorphism $p : F(L) \to E$ becomes zero both after composition with $F(K) \to F(L)$ and after composition with $F(A^{n_M}) \to F(L)$. However, since $K \oplus A^{n_M} \to L$ is surjective (algebra argument omitted), we conclude that $F(K \oplus A^{n_M}) \to F(L)$ is an epimorphism (by the above) whence $E = 0$. This finishes the proof. □